Elementary Analysis

Section 36: Improper Integrals

Exercises 36.1–36.15, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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36 Improper integrals

An improper integral is a *limit of ordinary integrals*; it is not an instruction to substitute an infinite number into an antiderivative. For example,

$$\int_a^b f = \lim_{d\to b^-} \int_a^d f$$

when the left endpoint is harmless. If both endpoints are troublesome, choose an interior point α and take the two limits separately. A finite limit means convergence. A limit $+\infty$ or $-\infty$ is an extended value, not convergence. An oscillating limit has no value. In particular, a symmetric limit on $[-R, R]$ cannot replace the two one-sided limits in the definition.

For Stieltjes integrals, F is increasing. Its increment $F(v) - F(u)$ replaces the interval length $v - u$. We use Ross's endpoint convention from Section 35, so that adjacent integrals add without counting a jump twice. A distribution function has total increment $F(\infty) - F(-\infty) = 1$.

36.1 Exercise 36.1: Agreement with an ordinary integral

What we must show. If f is Riemann integrable on the finite interval $[a, b]$, prove that $\lim_{d\to b^-} \int_a^d f = \int_a^b f$.

Plan. The only omitted contribution is over the short interval $[d, b]$.

Bound that contribution.

Riemann integrability includes boundedness. Choose $B \geq 0$ with $|f(x)| \leq B$ on $[a, b]$. Restrictions of an integrable function are integrable, and additivity and the integral inequality give

$$\left| \int_a^b f - \int_a^d f \right| = \left| \int_d^b f \right| \leq \int_d^b |f| \leq B(b - d).$$

If $B = 0$, the difference is already zero. If $B > 0$, given $\varepsilon > 0$, choosing $0 < b - d < \varepsilon/B$ makes the difference less than ε.

Conclusion. The difference tends to zero, so the two definitions agree.

36.2 Exercise 36.2: The splitting point does not change the value

What we must show. Show that the definition of an integral on (a, b) is independent of the chosen interior point.

Compare two splitting points by a finite integral.

Suppose $a < \alpha < \beta < b$ and set $C = \int_\alpha^\beta f$. This is a finite ordinary integral by the local integrability hypothesis. For $a < c < \alpha$ and $\beta < d < b$,

$$\int_c^\beta f = \int_c^\alpha f + C, \qquad \int_\beta^d f = \int_\alpha^d f - C.$$

Pass to the endpoint limits. Adding or subtracting a finite number preserves the existence and sign of an infinite limit. Thus, if $A = \int_a^\alpha f$ and $B = \int_\alpha^b f$ exist, then

$$\int_a^\beta f = A + C, \qquad \int_\beta^b f = B - C.$$

Check that the permitted sum is unchanged.

If A, B are finite, their sum is plainly unchanged. If one is infinite and their sum is permitted, the finite changes do not alter either infinite sign. An excluded sum $+\infty + (-\infty)$ remains excluded. The same argument in reverse proves the converse, and interchanging α, β covers either order.

Conclusion. The existence and the value of the improper integral do not depend on the split.

36.3 Exercise 36.3: Power singularities at zero and at infinity

What we must show. Analyze $\int_0^1 x^{-p}\,dx$ and $\int_0^\infty x^{-p}\,dx$ for $p > 0$.

(a) Cut off the singular endpoint before integrating.

For $0 < c < 1$ and $p \neq 1$, the power rule gives

$$\int_c^1 x^{-p}\,dx = \frac{1 - c^{1-p}}{1-p}.$$

If $0 < p < 1$, the exponent $1 - p$ is positive, so $c^{1-p} \to 0$; the answer is $1/(1-p)$. If $p > 1$, then $c^{1-p} = 1/c^{p-1} \to +\infty$. Rewriting the last display as $(c^{1-p} - 1)/(p-1)$ shows that the integral tends to $+\infty$. For completeness, the boundary case $p = 1$ has $\int_c^1 dx/x = -\log c \to +\infty$.

(b) Treat both ends separately.

For $p \neq 1$,

$$\int_1^d x^{-p}\,dx = \frac{d^{1-p} - 1}{1-p}.$$

This tends to $1/(p-1)$ if $p > 1$, and to $+\infty$ if $0 < p < 1$. For $p = 1$ it equals $\log d \to +\infty$. Thus $p > 1$ fails at zero, $0 < p < 1$ fails at infinity, and $p = 1$ fails at both ends. All integrands are nonnegative, so there is no cancellation issue.

Conclusion. The integral on $(0, 1)$ converges exactly for $0 < p < 1$; the integral on $(0, \infty)$ is $+\infty$ for every $p > 0$.

36.4 Exercise 36.4: Three explicit improper integrals

(a) The logarithm near zero.

Integration by parts on $[c, 1]$, where $c > 0$, yields

$$\int_c^1 \log x\,dx = [x \log x - x]_c^1 = -1 - c\log c + c.$$

To justify $c \log c \to 0$, put $t = 1/c$. The elementary estimate $\log t = \int_1^t du/u \le \int_1^t du/\sqrt{u} = 2(\sqrt{t} - 1)$ for $t \ge 1$ gives $0 \le (\log t)/t \le 2/\sqrt{t} \to 0$. Consequently $\int_0^1 \log x\,dx = -1$.

(b) A logarithm divided by its argument.

For $d > 2$, substitution $u = \log x$, $du = dx/x$, gives

$$\int_2^d \frac{\log x}{x}\,dx = \frac{(\log d)^2 - (\log 2)^2}{2} \longrightarrow +\infty.$$

(c) The reciprocal quadratic.

Since $(\arctan x)' = 1/(1+x^2)$,

$$\int_0^d \frac{dx}{1+x^2} = \arctan d \longrightarrow \frac{\pi}{2}.$$

Conclusion. The values, in order, are -1, $+\infty$, and $\pi/2$.

36.5 Exercise 36.5: A nonnegative integral as a supremum

What we must show. For continuous $f \geq 0$ on (a, b), identify its improper integral with the supremum of all integrals over compact subintervals.

Establish the two one-sided limits.

Choose $\alpha \in (a, b)$. Define

$$A(c) = \int_c^\alpha f \quad (a < c < \alpha), \qquad B(d) = \int_\alpha^d f \quad (\alpha < d < b).$$

Because $f \geq 0$, $A(c)$ increases as c moves toward a, and $B(d)$ increases as d moves toward b. The monotone-limit property therefore gives limits $A, B \in [0, \infty]$, equal to their respective suprema. The improper integral exists and equals $A + B$; this sum can never have the forbidden form $+\infty + (-\infty)$.

Compare with all compact subintervals.

Let $M = \sup\{\int_c^d f : [c, d] \subset (a, b)\}$. Enlarging any $[c, d]$ so that it contains α can only increase its integral. An enlarged interval has integral $A(c') + B(d') \leq A + B$; thus $M \leq A + B$. Conversely, all intervals with $c < \alpha < d$ are allowed in the supremum, so $M \geq A(c) + B(d)$. If A, B are finite, choose c, d making these terms within $\varepsilon/2$ of their suprema, and obtain $M \geq A + B - \varepsilon$. If either supremum is infinite, these same integrals exceed every prescribed bound, so $M = +\infty$.

Conclusion. $\displaystyle\int_a^b f = \sup_{[c,d]\subset(a,b)} \int_c^d f$, with $+\infty$ allowed.

36.6 Exercise 36.6: Comparison tests

What we must show. Suppose $0 \leq f \leq g$ and both functions are continuous on (a, b). Prove the convergence and divergence comparisons.

Compare finite intervals first.

For every compact $[c, d] \subset (a, b)$, ordinary integral monotonicity gives $0 \leq \int_c^d f \leq \int_c^d g$. Exercise 36.5 allows us to take suprema, giving

$$0 \leq \int_a^b f \leq \int_a^b g \leq +\infty.$$

(a) Finite upper bound.

If $\int_a^b g$ is finite, the middle quantity is bounded above by that finite number. The nonnegative integral of f therefore converges.

(b) Infinite lower bound.

If $\int_a^b f = +\infty$, no finite number can bound the integral of g from above. Therefore $\int_a^b g = +\infty$.

Conclusion. A smaller nonnegative function inherits convergence; a larger one inherits divergence to $+\infty$.

36.7 Exercise 36.7: The Gaussian integral

(a) Prove finiteness before attempting evaluation.

The integrand is even. On $[0,1]$ it is at most 1; on $[1,\infty)$, $x^2 \geq x$, so $e^{-x^2} \leq e^{-x}$. The comparison test gives

$$0 < I := \int_{-\infty}^{\infty} e^{-x^2}\,dx \leq 2\left(1 + \int_1^{\infty} e^{-x}\,dx\right) = 2(1+e^{-1}) < \infty.$$

(b) Explain the double-integral calculation.

This part uses polar coordinates for a double integral, as requested by the exercise. The following bounded-region argument avoids assuming in advance that improper iterated integrals may be interchanged. Set $I_R = \int_{-R}^{R} e^{-x^2}\,dx$. On the square $Q_R = [-R,R]^2$, integration of the continuous product gives

$$\iint_{Q_R} e^{-x^2-y^2}\,dx\,dy = I_R^2.$$

Let D_R be the disk of radius R. The inclusions $D_R \subset Q_R \subset D_{\sqrt{2}R}$ and positivity give a lower and an upper bound for I_R^2. In polar coordinates,

$$\iint_{D_R} e^{-x^2-y^2}\,dx\,dy = \int_0^{2\pi}\int_0^R e^{-r^2} r\,dr\,d\theta = \pi(1-e^{-R^2}).$$

Here the extra factor r is the polar area factor, and $\int_0^R e^{-r^2} r\,dr = (1-e^{-R^2})/2$. Hence

$$\pi(1-e^{-R^2}) \leq I_R^2 \leq \pi(1-e^{-2R^2}).$$

Both outer expressions tend to π. Since part (a) shows $I_R \to I$, we obtain $I^2 = \pi$; and $I > 0$ selects the positive root.

Conclusion. $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.

36.8 Exercise 36.8: Absolute convergence implies convergence

What we must show. If f is continuous on (a,b) and $\int_a^b |f| < \infty$, prove that $\int_a^b f$ exists and is finite.

Plan. Write f as the difference of two nonnegative functions whose integrals are bounded above by integrals already known to converge.

Use positive and negative parts.

Define

$$f_+(x) = \max\{f(x), 0\}, \qquad f_-(x) = \max\{-f(x), 0\}.$$

These are continuous because, for example, $f_+ = (|f| + f)/2$. Pointwise, $0 \le f_+, f_- \le |f|$ and $f = f_+ - f_-$. Exercise 36.6 proves that both nonnegative improper integrals are finite.

Pass to limits without subtracting infinities.

On each compact subinterval, ordinary linearity gives $\int_c^d f = \int_c^d f_+ - \int_c^d f_-$. Fix an interior split, and take the left and right endpoint limits separately. Each term has a finite limit, so the limit law for subtraction applies.

Conclusion. $\int_a^b f$ converges and $\left|\int_a^b f\right| \le \int_a^b |f|$.

36.9 Exercise 36.9: Integrals for the normal distribution

Write $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and $F(t) = \int_{-\infty}^t \varphi(x)\,dx$. Substitution $x = \sqrt{2}u$ in Exercise 36.7 gives $\int_{\mathbb{R}} \varphi = 1$.

(a) Replace the Stieltjes integrator by its density.

The Fundamental Theorem of Calculus gives $F' = \varphi$, a continuous function. For continuous f, Theorem 35.13 on each finite $[a, b]$ therefore yields

$$\int_a^b f\,dF = \int_a^b f(x)F'(x)\,dx = \frac{1}{\sqrt{2\pi}} \int_a^b f(x)e^{-x^2/2}\,dx.$$

Taking the two endpoint limits separately proves the same identity on $\mathbb{R}$ whenever the integral specified in the question exists; positive constant multiplication preserves finite and infinite limits.

(b) The second moment.

Since $\varphi'(x) = -x\varphi(x)$, integration by parts on $[a, b]$ gives

$$\int_a^b x^2\varphi(x)\,dx = [-x\varphi(x)]_a^b + \int_a^b \varphi(x)\,dx.$$

The boundary term tends to zero at either infinity. For instance, $e^t \ge t^2/2$ for $t \ge 0$ implies $xe^{-x^2/2} \le 8/x^3$ for $x > 0$, and negative x are handled by absolute values. Thus $\int_{\mathbb{R}} x^2\,dF = 1$.

(c) The exponential of the square.

The integrand here is e^{x^2}, not e^x. Its density product is

$$e^{x^2}\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{x^2/2} \ge \frac{1}{\sqrt{2\pi}}.$$

The integral of this positive constant over an infinite half-line is infinite, so comparison proves $\int_{\mathbb{R}} e^{x^2}\,dF = +\infty$.

(d) The absolute first moment.

Evenness and the antiderivative of $xe^{-x^2/2}$ give

$$\int_{\mathbb{R}} |x|\,dF = \frac{2}{\sqrt{2\pi}} \int_0^\infty xe^{-x^2/2}\,dx = \frac{2}{\sqrt{2\pi}}[-e^{-x^2/2}]_0^\infty = \sqrt{\frac{2}{\pi}}.$$

(e) The first moment and a necessary convergence check.

Part (d) shows absolute convergence. Each half-integral is therefore finite. Because $x\varphi(x)$ is odd, substitution $u = -x$ in the negative half-integral makes it the negative of the positive half. Their sum is zero. Oddness alone would not suffice without the existence of both halves.

Conclusion. The values in (b)–(e) are 1, $+\infty$, $\sqrt{2/\pi}$, and 0.

36.10 Exercise 36.10: Linearity on the whole real line

What we must show. For F-integrable f, g on $\mathbb{R}$, prove integrability and the sum rule for $f + g$.

Apply finite-interval linearity.

For every finite $a < 0 < b$, Theorem 35.8 gives

$$\int_a^0 (f+g)\,dF = \int_a^0 f\,dF + \int_a^0 g\,dF, \qquad \int_0^b (f+g)\,dF = \int_0^b f\,dF + \int_0^b g\,dF.$$

Since f, g are F-integrable on $\mathbb{R}$, all four half-line limits are finite. The limit law for addition proves that the two limits for $f + g$ exist and are finite. Add them and regroup finite real numbers.

Conclusion. $f + g$ is F-integrable and $\int_{\mathbb{R}}(f+g)\,dF = \int_{\mathbb{R}} f\,dF + \int_{\mathbb{R}} g\,dF$. The same argument proves scalar multiplication, including multiplication by -1.

36.11 Exercise 36.11: Monotonicity on the whole real line

What we must show. If f, g are F-integrable and $f \le g$, prove the integral inequality.

Preserve the inequality through finite limits.

Theorem 35.9 gives $\int_a^0 f\,dF \le \int_a^0 g\,dF$ for each $a < 0$, and $\int_0^b f\,dF \le \int_0^b g\,dF$ for each $b > 0$. Both sides have finite limits by hypothesis. If real-valued quantities satisfy an inequality before taking a limit, the limits satisfy the same weak inequality. Pass to $a \to -\infty$ and $b \to +\infty$ separately, and add.

Conclusion. $\displaystyle\int_{\mathbb{R}} f\,dF \le \int_{\mathbb{R}} g\,dF.$

36.12 Exercise 36.12: Comparison for improper Stieltjes integrals

What we must show. State and prove the analogue of Exercise 36.6. Assume F is increasing and f, g are F-integrable on every finite interval, with $0 \le f \le g$. Continuity of f, g is one sufficient condition for local integrability.

Nonnegativity supplies the endpoint limits.

For $b > 0$, $\int_0^b f\,dF$ is increasing in b, since its increment over an added interval is nonnegative. Likewise, $\int_a^0 f\,dF$ increases as $a \to -\infty$. Thus both limits exist in $[0, \infty]$. The same holds for g.

Compare before and after the limits.

Finite-interval monotonicity gives

$$0 \le \int_a^0 f\,dF \le \int_a^0 g\,dF, \qquad 0 \le \int_0^b f\,dF \le \int_0^b g\,dF.$$

Taking limits and adding nonnegative extended values proves $0 \le \int_{\mathbb{R}} f\,dF \le \int_{\mathbb{R}} g\,dF$.

Conclusion. If $\int_{\mathbb{R}} g\,dF < \infty$, then $\int_{\mathbb{R}} f\,dF < \infty$. If $\int_{\mathbb{R}} f\,dF = +\infty$, then $\int_{\mathbb{R}} g\,dF = +\infty$.

36.13 Exercise 36.13: Absolute convergence for Stieltjes integrals

What we must show. If f is continuous on $\mathbb{R}$ and $\int_{\mathbb{R}} |f|\,dF < \infty$, prove f is F-integrable on $\mathbb{R}$.

Obtain finite nonnegative integrals.

Continuity makes f, $|f|$, and $h = f + |f|$ locally F-integrable. Pointwise, $0 \le h \le 2|f|$. Exercise 36.12 shows that h has a finite integral, since $2|f|$ does.

Subtract a function with a known finite integral.

Scalar multiplication shows $-|f|$ is F-integrable. Exercise 36.10 now applies to $f = h + (-|f|)$, so f is F-integrable. Applying monotonicity to $-|f| \le f \le |f|$ also gives $|\int_{\mathbb{R}} f\,dF| \le \int_{\mathbb{R}} |f|\,dF$.

Conclusion. Absolute F-integrability implies F-integrability. The argument still works if continuity is replaced by local F-integrability of f and $|f|$.

36.14 Exercise 36.14: A countable collection of point masses

What we must show. Let the distinct points be $u_j \in \mathbb{R}$, let $c_j \ge 0$, and suppose $C = \sum_{j=1}^{\infty} c_j < \infty$. Analyze $F(t) = \sum_j c_j J_{u_j}(t)$, where $J_u(t) = 0$ for $t < u$ and $J_u(t) = 1$ for $t \ge u$ on $\mathbb{R}$.

(a) Existence and monotonicity.

For each t, $0 \le c_j J_{u_j}(t) \le c_j$. The series therefore converges, and its tail is at most $\sum_{j>N} c_j$, independently of t. Thus convergence is uniform. Each summand is increasing; for $s < t$, termwise comparison gives $F(s) \le F(t)$.

(b) Finite intervals and their endpoint convention.

Let $|f| \le B$ on $\mathbb{R}$. Theorem 35.12(b), or its finite-mass tail estimate, makes every bounded f F-integrable on $[a, b]$. For this right-continuous F, the formula is

$$\int_a^b f\,dF = \sum_{j:\, a<u_j\le b} c_j f(u_j).$$

Indeed a single global step contributes its mass exactly when $a < u_j \le b$; a jump at a has already occurred before the interval's increment begins. Summing the steps is justified because the absolute value of the contribution from all indices $j > N$ is at most $B \sum_{j>N} c_j$. This tends to zero.

Pass to the whole line, controlling the tails.

The numerical series $\sum_j c_j f(u_j)$ is absolutely convergent, because $\sum_j |c_j f(u_j)| \le BC$. To see directly that the finite-interval sums approach it, choose N with $B \sum_{j>N} c_j < \varepsilon$ (the case $B = 0$ is immediate), then choose $a < \min_{j\le N} u_j$ and $b > \max_{j\le N} u_j$. All first N terms are included, and the omitted terms have total absolute value below ε. The same argument on the two halves treats $u_j = 0$ in the negative half only. Hence

$$\int_{-\infty}^{\infty} f\,dF = \sum_{j=1}^{\infty} c_j f(u_j).$$

(c) The total mass condition.

The same finite-head/small-tail argument gives $F(-\infty) = 0$ and $F(\infty) = C$. Consequently F is a distribution function exactly when $\sum_j c_j = 1$. It is also right-continuous: each finite partial sum is right-continuous, and uniform convergence preserves this one-sided continuity.

(d) Dense jumps and a necessary qualification.

For $s < t$,

$$F(t) - F(s) = \sum_{j:\, s<u_j\le t} c_j.$$

If the u_j enumerate $\mathbb{Q}$ and *every* $c_j > 0$, there is a rational u_j strictly between s and t, so this difference is positive. Thus F is strictly increasing. More generally the exact condition is that $\{u_j : c_j > 0\}$ is dense in $\mathbb{R}$.

Conclusion. Parts (a)–(c) hold for nonnegative coefficients as stated. Part (d), read with merely nonnegative coefficients, needs the additional density condition: all $c_j = 0$ would give $F = 0$. This is the positive-coefficient situation of Exercise 35.12.

36.15 Exercise 36.15: Uniform convergence on spaces of infinite and finite mass

(a) Small height can be offset by large width.

Define $f_n(x) = 1/n$ on $[0, n]$, and $f_n(x) = 0$ elsewhere. Each function is piecewise constant and has a finite improper Riemann integral. Its rectangle has height $1/n$ and width n, so

$$\int_{\mathbb{R}} f_n(x)\,dx = 1, \qquad \sup_{x\in\mathbb{R}} |f_n(x) - 0| = \frac{1}{n} \longrightarrow 0.$$

Thus uniform convergence alone does not preserve an integral over a domain of infinite ordinary length.

(b) Local integrability of the uniform limit.

Now suppose F is a distribution function, all f_n are F-integrable on $\mathbb{R}$, and $f_n \to f$ uniformly on $\mathbb{R}$. On each finite $[a, b]$, the uniform-limit theorem for Stieltjes integrals (Exercise 35.6) makes f F-integrable there. For clarity, that theorem follows from

$$U_F(f, P) - L_F(f, P) \le U_F(f_n, P) - L_F(f_n, P) + 2\|f - f_n\|_\infty[F(b) - F(a)] :$$

first choose n to make the second term small, then choose P using integrability of f_n.

Prove global integrability without assuming global boundedness.

Choose one index N with $\|f - f_N\|_\infty \le 1$. The difference $h = f - f_N$ is bounded and locally F-integrable. Since F has finite total increment, Theorem 36.4 makes h F-integrable on $\mathbb{R}$. Exercise 36.10 now shows that $f = f_N + h$ is F-integrable. This argument does not require f_N, or f, to be bounded on all of $\mathbb{R}$.

Estimate the difference of the integrals.

For any sufficiently large n, put $\eta_n = \|f - f_n\|_\infty$. By monotonicity and the total-mass identity $\int_{\mathbb{R}} 1\,dF = F(\infty) - F(-\infty) = 1$,

$$-\eta_n \le \int_{\mathbb{R}} (f - f_n)\,dF \le \eta_n.$$

Linearity therefore gives

$$\left| \int_{\mathbb{R}} f \, dF - \int_{\mathbb{R}} f_n \, dF \right| \leq \eta_n \longrightarrow 0.$$

Conclusion. For a distribution function, uniform convergence does preserve the integral. The decisive bound is the finite total mass 1, rather than the geometric length of the real line.